National College of Ireland

BSc. (Hons) in Computing (Daytime), Year 2, BSHCIFSC2, BSHCIFSC2_AE_SD

Higher Certificate in Computing (Daytime), Year 2, HCCOMP2

BSc. (Hons) in Computing (Evening), Year 2, BSHCE2

Higher Certificate in Computing (Evening), Year 2, HCCOMPE2

Release: 5th March, submit: 18th March 2026

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Data Structures & Algorithms CA1

Smart Docklands

Mobility Innovation Project Proposal

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Introduction

For my Data Structures & Algorithms CA, we are tasked to build an application that solves a problem in relation to one of the four Smart Dockland Pilot Call themes. The 4 themes that are available are:

- **Urban Greening/Biodiversity**
- **Mobility Innovation**
- **Accessibility/ Inclusive Design**
- **Digital Tools / Tech Access**

The theme that I opted to go for was **Mobility Innovation** as I believe improving our different modes of transport helps to promote sustainability and it would be collectively beneficial to the community to entice people to use greener methods of commuting.

The Problem

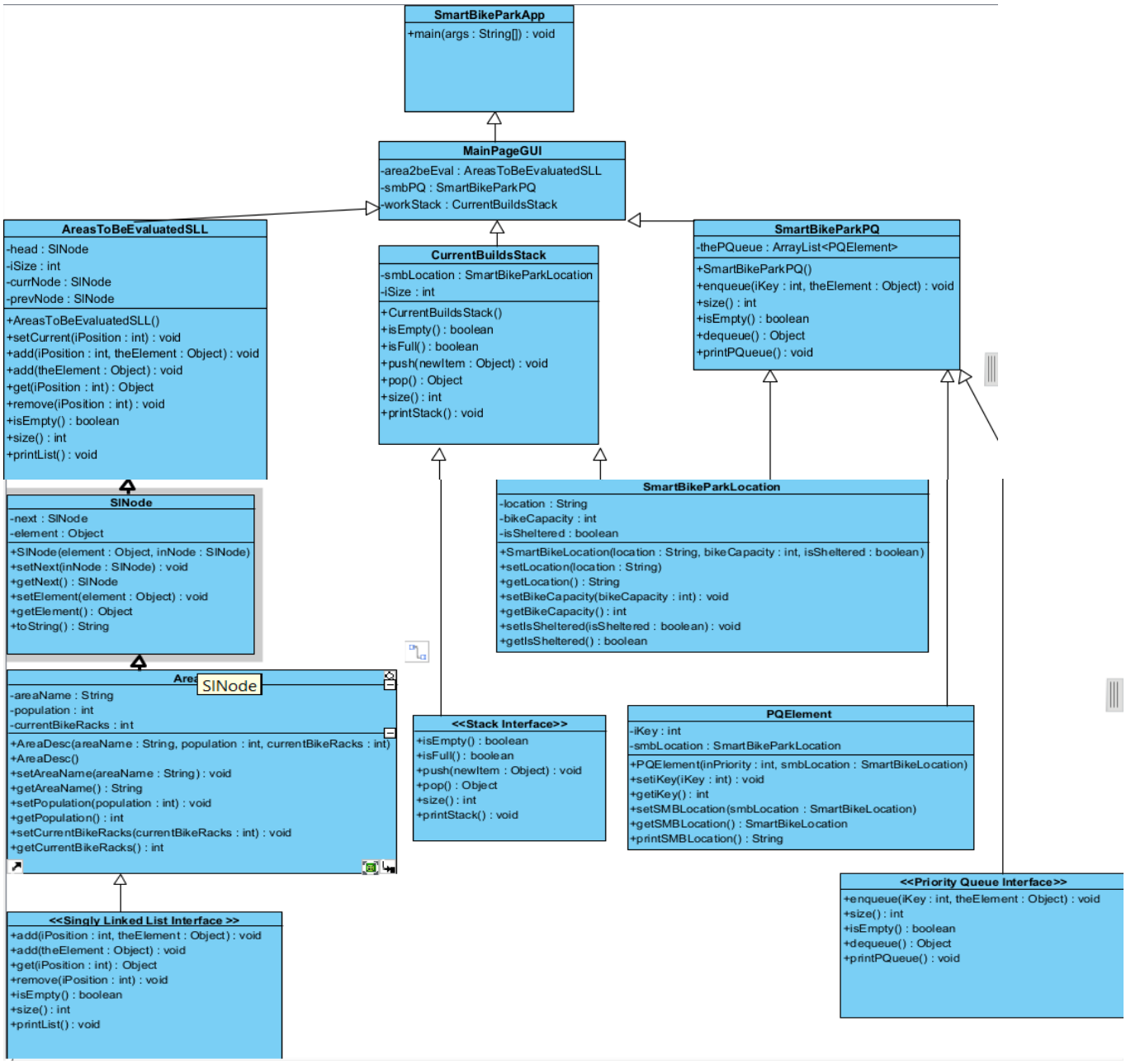
Travelling to college everyday can be expensive toward your own pocket and to the cities carbon footprint. In the last year I have decided to cycle to college as it is only a 25km up and back. A small inconvenience as a cycle in the cold or the rain is insignificant when trying to be more sustainable. As somebody who cycles with an e-bike into the college every day I have always noticed that there are no facilities to safely lock and charge your bike at the same time. Luckily for me my battery just about lasts the journey up and back whereas other people may not be as fortunate with their battery. I believe if we start implementing **Smart Bike Parking Systems** in the docklands it will promote people to make that extra effort to be more sustainable and cycle into town using their own E-Bike, knowing that they can lock their bike securely and charge it for their journey home.

The Solution

The app I am designing will first inform users about the Smart Bike Parking systems and how it can benefit their community. Afterwards the user will have the ability to add an area in the Docklands where they believe one of systems will be beneficial. This will be on a priority basis; users can determine which area is of higher priority. Afterwards the user can add from the priority queue to a queue of 3-5 jobs that will be started immediately. Due to the future success of this program in the docklands, the last feature will include where users can add future areas in Dublin to be evaluated for implementation of this programme. This app and problem that it solves fits very well with the theme chosen as it is innovating mobility.

> *"Pilot and showcase new mobility solutions in the Docklands, combining emerging technologies with sustainable, people-centred transport, and measure uptake and equity."*

Class Diagram



GitHub Link

https://github.com/x24528929/DSA-CA1-Smart-E-Bike-Docking-Station-App